Page No
Date

Pillars of OOPS 1. Access Modifier

* Access Modifier

In Java there four Access Modifiers
↳ Public
↳ Default
↳ Protected
↳ Private

Access modifiers are
↳ who have access to
- variables
- Methods
- constructor
- class

1. ~~Private~~ Let's take example of Bank which contains data such as balance, IFSC code etc. To restrict access to balance we can use Access Modifier.

1. Private
↳ specified using private keyword.
↳ The methods or data members declared as private are accessible only within class in which they are declared.

eg:

```
class BankAccount {
    private double balance;
    or
    private void withdraw() {
        balance--;
    }
}
```

* Package group similar classes / Interface together.

2. Default
↳ If no access modifier is specified, the member has default access and can only be accessed within same package.
↳ Anyone can access belonging to same package.

eg:

```
class car {
    String model;    // default access.
}
public class main {
    public static void main (String[] args)
    {
        car c = new car();
        c.model = "Tesla";    // accessible within same package.
        S.o.Pln (c.model);
    }
```

3. Protected.
↳ specified using protected keyword.
↳ accessible within same package or subclass in diff package. (child class)

4. Public
↳ specified using public keyword
↳ public members are accessible from everywhere. Any on ~~class~~ can access.

Private	default	protected	Public

Page No.
Date

Four Pillars of oops
↳ 1) Encapsulation
2) Inheritance
3) polymorphism
4) Abstraction

1) Encapsulation
↳ wrapping data and method together into a single unit.

Encapsulation says:
↳ ① Both data & Behaviour should be together encapsulated within a object
② we should not provide unrestricted access to data. (use access modifier)

eg: class Animal
{
int no. of legs } data
int type
void eat() {
} } method
}

To acheive encapsulation in Java
↳ Declare variables as Private
↳ Declare setters & getters to set and get variable values